Yong Lin

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Immunologist with expertise in T cell development, autoimmune diseases, and immune tolerance mechanisms. Experienced in high-parameter flow cytometry, immune cell assays, immunofluorescence microscopy, and various molecular biology techniques. Gained computational and Claude AI-assisted data analysis skills in recent research projects. Adept at resolving scientific and technical challenges to move projects forward.

Positions

2022-now **Postdoctoral Fellow, Hannah Meyer Lab, Cold Spring Harbor Laboratory**

- Developed and validated a novel enrichment and FACS sorting protocol for rare human thymic epithelial cells (TECs), enabling reproducible downstream scRNAseq characterization and resolving discrepancies in the field regarding cell surface markers used for the identification of TEC subsets.
- Established an essential markers flow cytometry panel and gating strategy for the identification of major human thymic epithelial cell subsets.
- Designed 10+ color flow cytometry panels for use on BD Fortessa and Sony ID7000 flow cytometers.
- Implemented and optimized protocols for the isolation and stimulation of human T cells and monocytes.
- Led a team of 3 technicians and 2 graduate students in a large-scale human thymus biobanking initiative, coordinating tissue collection across multiple sites in support of a multi-year human T cell development study.

2016-2022 **Graduate Research Assistant, Matthew Bettini Lab, Baylor College of Medicine**

- Designed and executed mechanistic studies in the NOD mouse model of autoimmune diabetes, demonstrating that neonatal antibody-mediated delivery of a Hybrid Insulin Peptide (HIP) autoantigen to thymic Langerin+ antigen-presenting cells enhances T cell central tolerance and delays autoimmune diabetes onset.
- Designed multi-parameter (8+ colors) flow cytometry panels for characterizing T cell developmental markers.
- Performed T cell functional assays such as T cell stimulation, proliferation, and cytokine production.
- Mentored 6 graduate students and 3 lab technicians in various lab techniques such as flow cytometry, animal handling, and mammalian cell culture.

2011-2016 **Research Technician, University of Virginia and Roskamp Institute**

- Performed *in vitro* and *in vivo* assays, including mammalian cell culture, laboratory animal injections, fluorescence microscopy, western blot, and ELISA, that contributed data to five peer-reviewed publications.
- Implemented and optimized new assays and standard operating procedures.
- Supervised new personnel in the lab.

Education

2016-2022 Ph.D. student, Bettini Lab, Baylor College of Medicine and University of Utah

2008-2012 B.S., University of Virginia School of Engineering

Laboratory Skills

- Flow Cytometry and FACS Cell Sorting (regularly perform 8+ color flow panels)
  - BD LSR Fortessa, Sony ID7000, Sony SH800, Sony MA900, BD FACSAria
- Single-Cell RNA sequencing sample preparation
  - Parse Biosciences platform, 10x Genomics Flex platform
- Molecular & Cellular Biology

    - DNA/RNA isolation, PCR, RT-qPCR, Western Blot, ELISA, histology (H&E, IHC, ICC), brightfield microscopy, fluorescence microscopy (Zeiss Observer), confocal microscopy (Zeiss LSM710 and LSM910), slide scanner (Zeiss Axioscan, Olympus VS200)
- In Vitro Assays
    - Mouse and human tissue processing for single cell analysis, mammalian cell culture, PBMC isolation, primary T cell isolation, primary monocyte isolation, immunoassays, T cell stimulation assays, monocyte stimulation assays, cell lines (Jurkat, THP-1, Neuronal cell lines, HEK293)
- In Vivo Models
    - NOD mouse model of autoimmune diabetes, laboratory mouse handling, surgical techniques (mouse thymus transplant), IV/IP/SC/ID injections
- Bioinformatics & Data Analysis
    - Excel, GraphPad Prism, QuantStudio, ImageJ, QuPath, FlowJo, R Programming (Seurat), scRNAseq data analysis (clustering, UMAP, differential gene expression, cell type annotation), Claude AI-assisted computational data analysis in R

Select Honors & Presentations

2023 Invited speaker on sample preparation using Parse Biosciences scRNAseq platform
2022 Oral presentation at St. Jude National Graduate Student Symposium
2021 Abstract Award and oral presentation at American Association of Immunologists Annual Meeting
2019 1st place, Baylor College of Medicine Alumni Reunion Research Poster Showcase
2018 Travel award and oral presentation at American Association of Immunologists Annual Meeting

First Author Publications (peer-reviewed, pre-prints, and manuscript-in-preparations)

1. **Lin Y**, Lapine M, Netz D, Chapin SR, Prabakar RK, Meyer HV. Isolation and Characterization of Human Thymic Epithelial Cells. (Journal of Immunology, accepted 9/23/2026)
    - Established and standardized a novel protocol for the consistent identification of major human thymic epithelial cell subsets and resolved discrepancies in the field regarding thymic epithelial cell surface markers.
2. **Lin Y**, Perovanovic J, Kong Y, Igyarto BZ, Zurawski S, Tantin D, Zurawski G, Bettini M, Bettini ML. Antibody-Mediated Targeting of a Hybrid Insulin Peptide Toward Neonatal Thymic Langerin-Positive Cells Enhances T-Cell Central Tolerance and Delays Autoimmune Diabetes. Diabetes. 2022 Aug 1;71(8):1735-1745. doi: 10.2337/db21-1069. PMID: 35622068.
    - Demonstrated HIP-targeted tolerance induction delays autoimmune diabetes onset in the NOD mouse model

Full list of publications available at https://www.ncbi.nlm.nih.gov/myncbi/1ZaIsBozFZEA1/bibliography/public/